File Handling Programming Questions

1. Write a program to create a file named 'data.txt' and write 5 lines of text into it.
   Then display the contents of the file.
2. Write a program to read a file and count the total number of lines present in it.
3. Write a program to read a file and count the total number of words in it.
4. Write a program to count the total number of characters in a file excluding spaces.
5. Write a program to search for a particular word in a file and display how many times it appears.
6. Write a program to copy the contents of one file into another file.
7. Write a program to append new data to an existing file and display the updated contents.
8. Write a program to read a file and count:
   - Number of vowels
   - Number of consonants
   - Number of digits
   - Number of special characters
9. Write a program to find total number of prime numbers of a given file and store the result in another file
10. Write a program to find calculation as mentioned below on subjectmarks.txt (View the given text file before start)
    a. Create another file with StudentID and TotalMarks
    b. Find topper Student with Marks
    c. Display the Total Students Passed, Total Students Fail and Total Number of Students

Q1. What is the output?

```
# test.txt contains: "Learning Python is Fun"
f = open("test.txt", "r")
print(f.read(8))
print(f.read(6))
f.close()
```

Q2. What will be printed?

```
# lines.txt contains:
# One
# Two
# Three
f = open("lines.txt", "r")
L = f.readlines()
print(len(L))
f.close()
```

Q3. Predict the output:

```
# data.txt contains: "1234567890"
with open("data.txt", "r") as f:
    val1 = f.read(3)
    f.seek(0)
    val2 = f.read(2)
    print(val1 + val2)
```

Q4. What is the final content of `out.txt`?

```
f = open("out.txt", "w")
f.write("Alpha")
f.close()

f = open("out.txt", "a")
f.write("Beta")
f.close()

f = open("out.txt", "w")
f.write("Gamma")
f.close()
```

Q5. Identify the error or output:

```
f = open("num.txt", "w")
f.write(100)
f.close()
```

Q6. What will be the output of the following?

```
# msg.txt is empty
with open("msg.txt", "w+") as f:
    f.write("Python")
    f.seek(0)
    print(f.read(2))
```

Q7. What is the output of `tell()`?

```
# story.txt contains: "Once upon a time"
f = open("story.txt", "r")
f.readline()
print(f.tell())
f.close()
```

Q8. Predict the output (Binary Mode):

```
# bin.dat contains: b"ABCDE"
with open("bin.dat", "rb") as f:
    f.seek(2)
    f.seek(1, 1) # Note: 1 is 'cur' position
    print(f.read(1))
```

Q9. What happens in this loop?

```
# colors.txt contains:
# Red
# Blue
f = open("colors.txt", "r")
for line in f:
    print(line.strip().upper(), end="#")
f.close()
```

Q10. How many characters are read?

```
# temp.txt contains: "Hello\nWorld"
f = open("temp.txt", "r")
data = f.read().split()
print(len(data))
f.close()
```

Keys

Q#	Expected Output / Result	Reason
1	Learning Pytho	Note the space before "Pytho".
2	3	readlines() creates a list of 3 string elements.
3	12312	seek(0) resets the pointer to the start.
4	Gamma	The final "w" mode wiped "AlphaBeta".
5	TypeError	write() only accepts strings, not integers.
6	Py	w+ allows reading after writing, but you must seek(0).
7	15	It returns the total bytes of the first line (including \n if present).
8	b'D'	Move to 2, then move 1 more forward = position 3 ('D').
9	RED#BLUE#	strip() removes newlines; end="# replaces the print newline.
10	2	split() breaks "Hello\nWorld" into a list: ['Hello', 'World'].